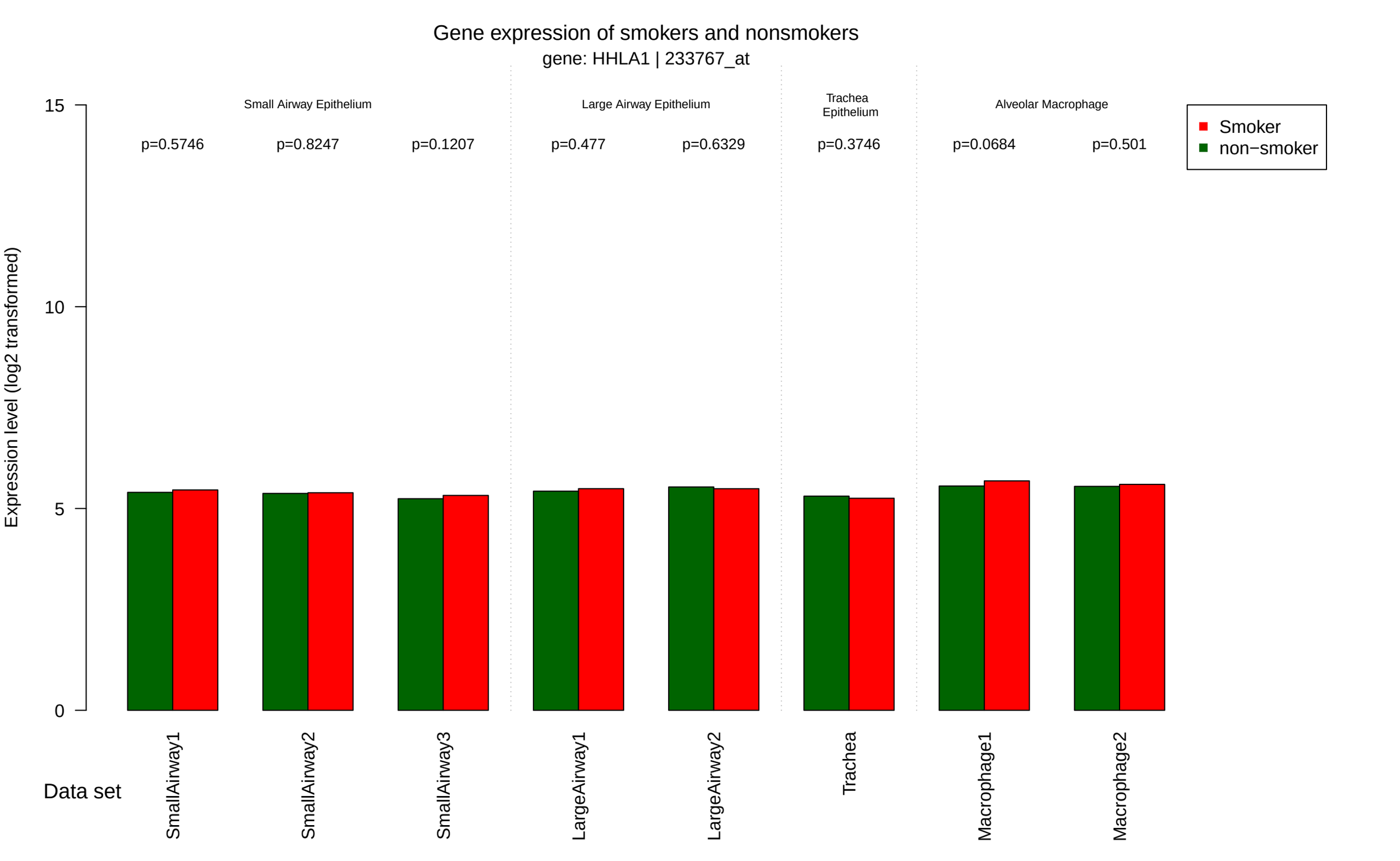
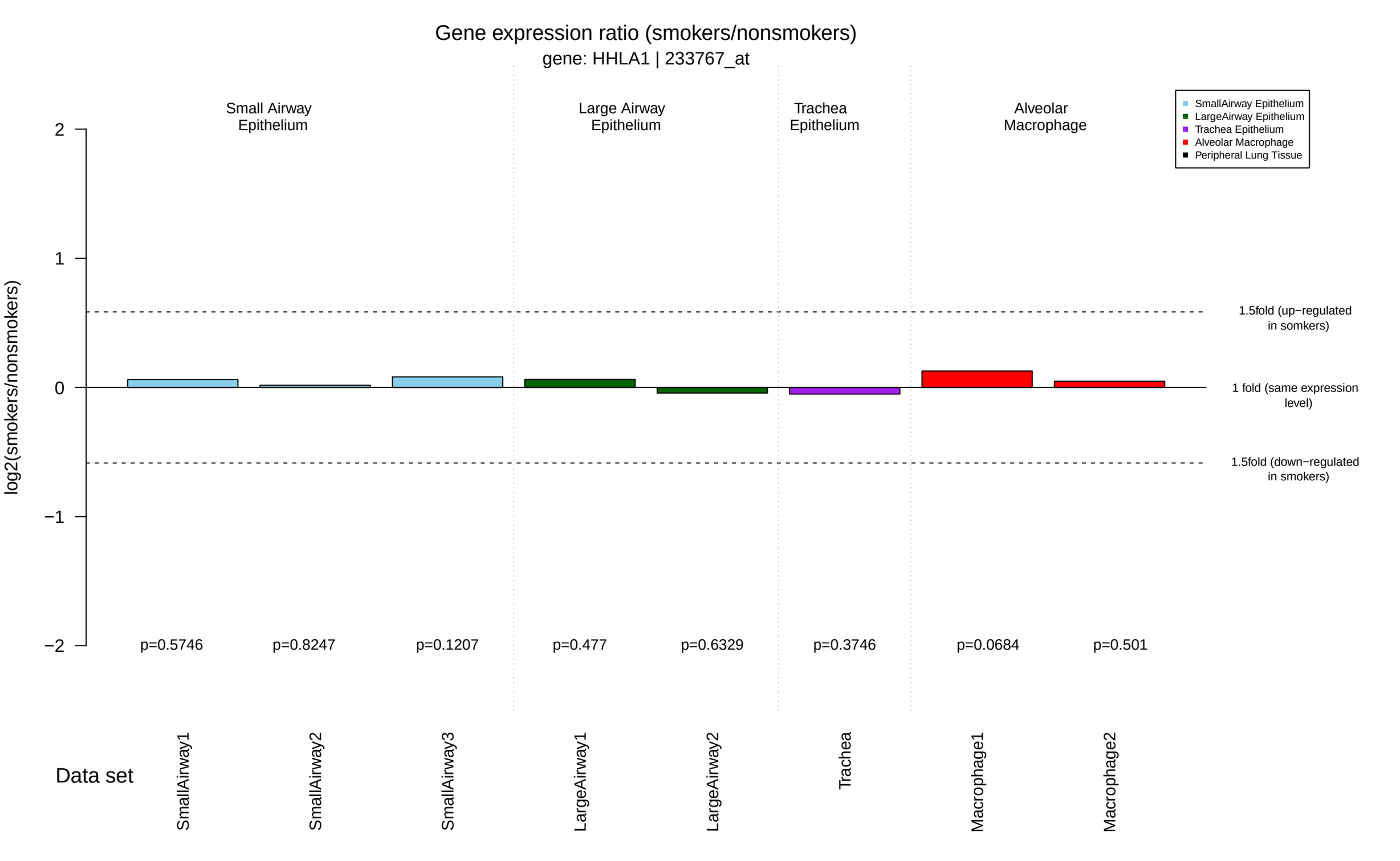
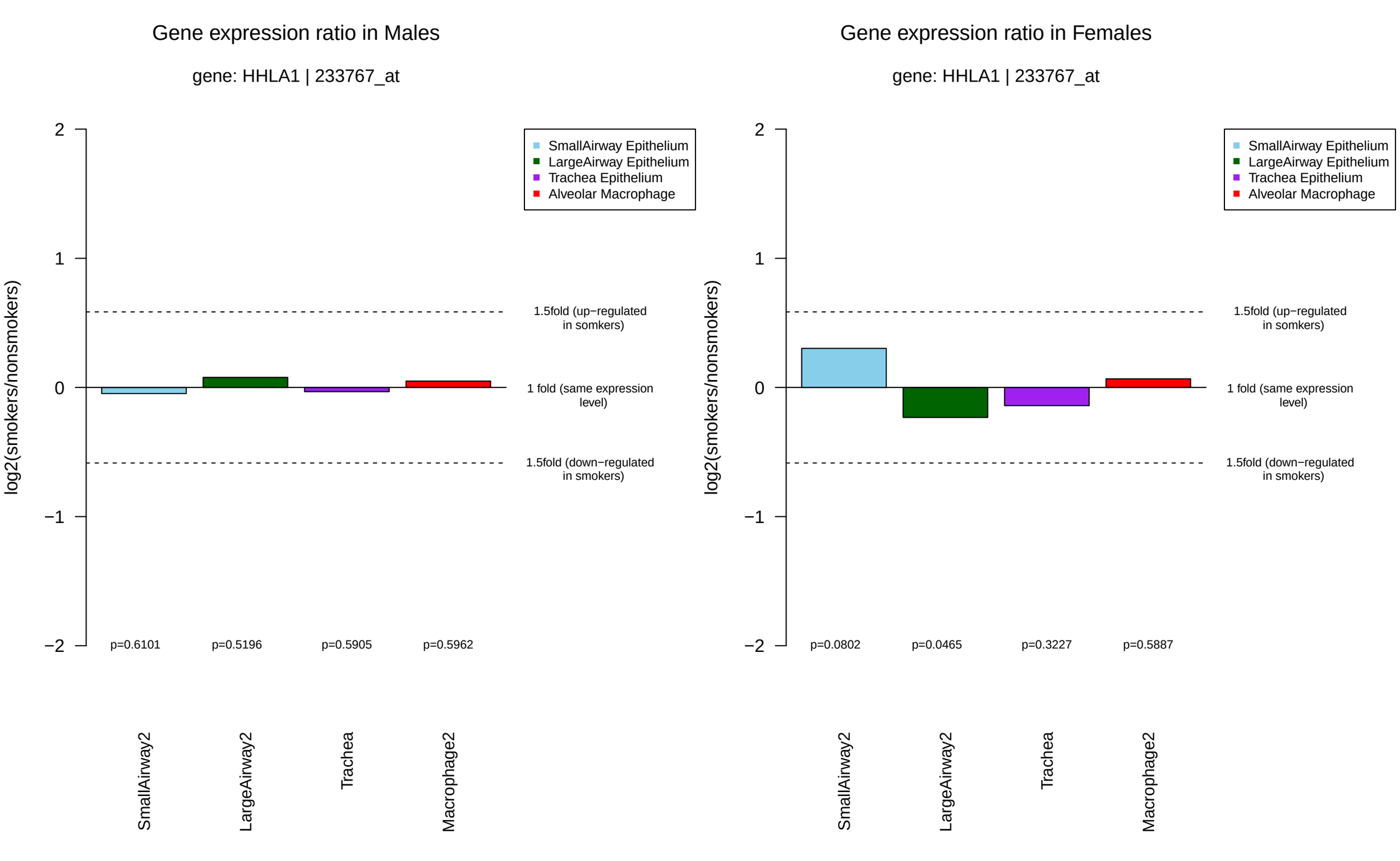
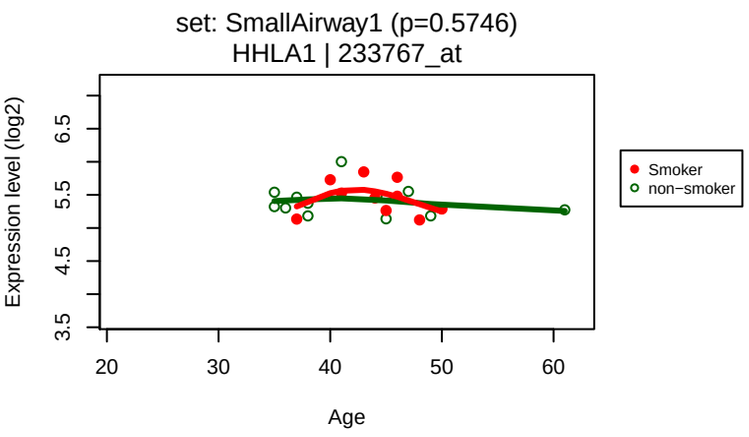
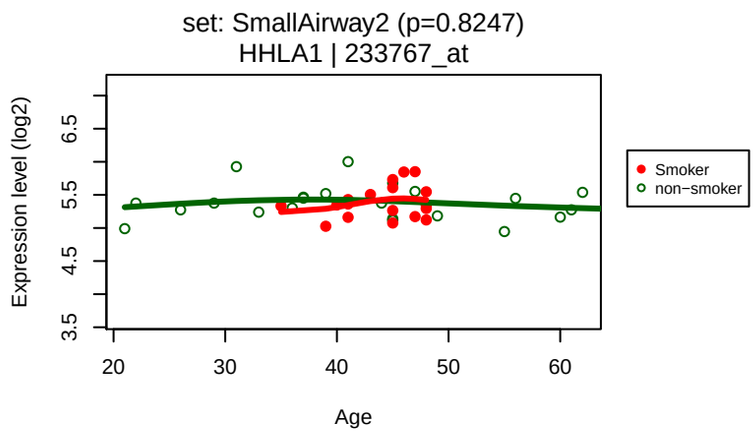
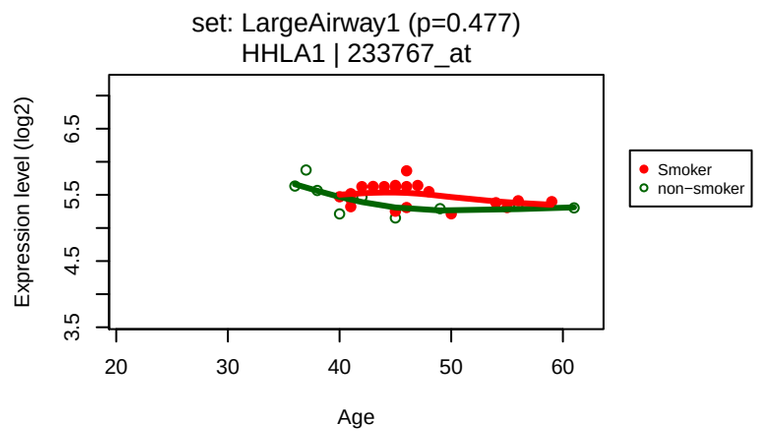

Gene expression of smokers and nonsmokers
gene: HHLA1 | 233767_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5746
p=0.8247
p=0.1207
p=0.477
p=0.6329
p=0.3746
p=0.0684
p=0.501
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

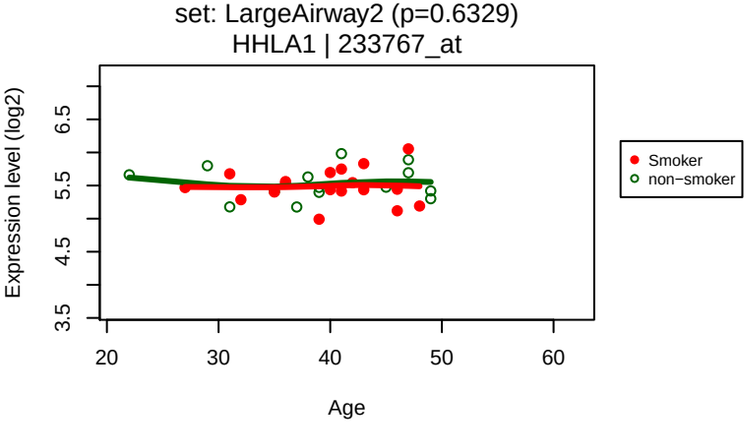

Gene expression ratio (smokers/nonsmokers)
gene: HHLA1 | 233767_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5746
p=0.8247
p=0.1207
p=0.477
p=0.6329
p=0.3746
p=0.0684
p=0.501
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

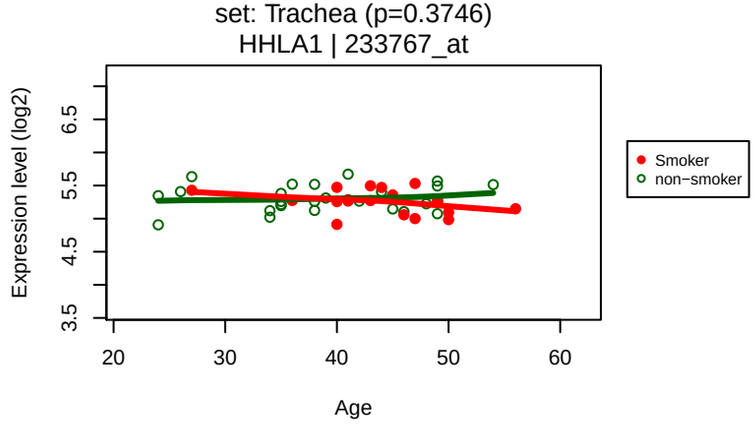

Gene expression ratio in Males
gene: HHLA1 | 233767_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6101
p=0.5196
p=0.5905
p=0.5962
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HHLA1 | 233767_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0802
p=0.0465
p=0.3227
p=0.5887
SmallAirway2
LargeAirway2
Trachea
Macrophage2

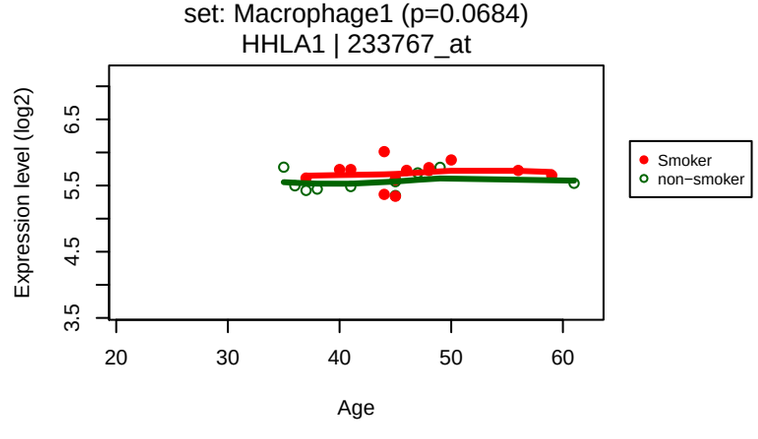

set: SmallAirway1 (p=0.5746)
HHLA1 | 233767_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker

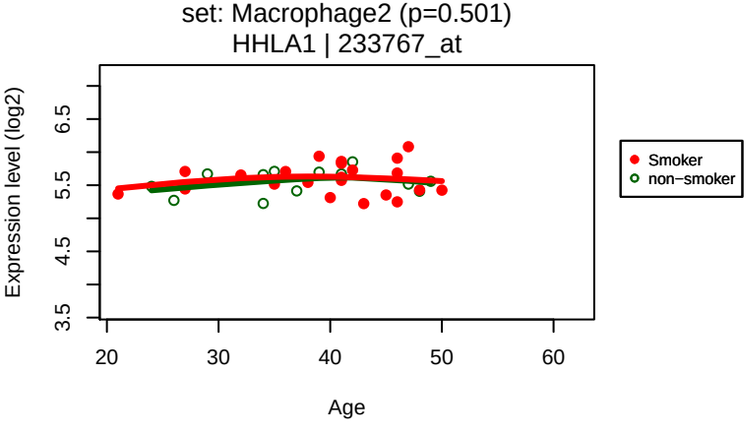

set: SmallAirway2 (p=0.8247)
HHLA1 | 233767_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker

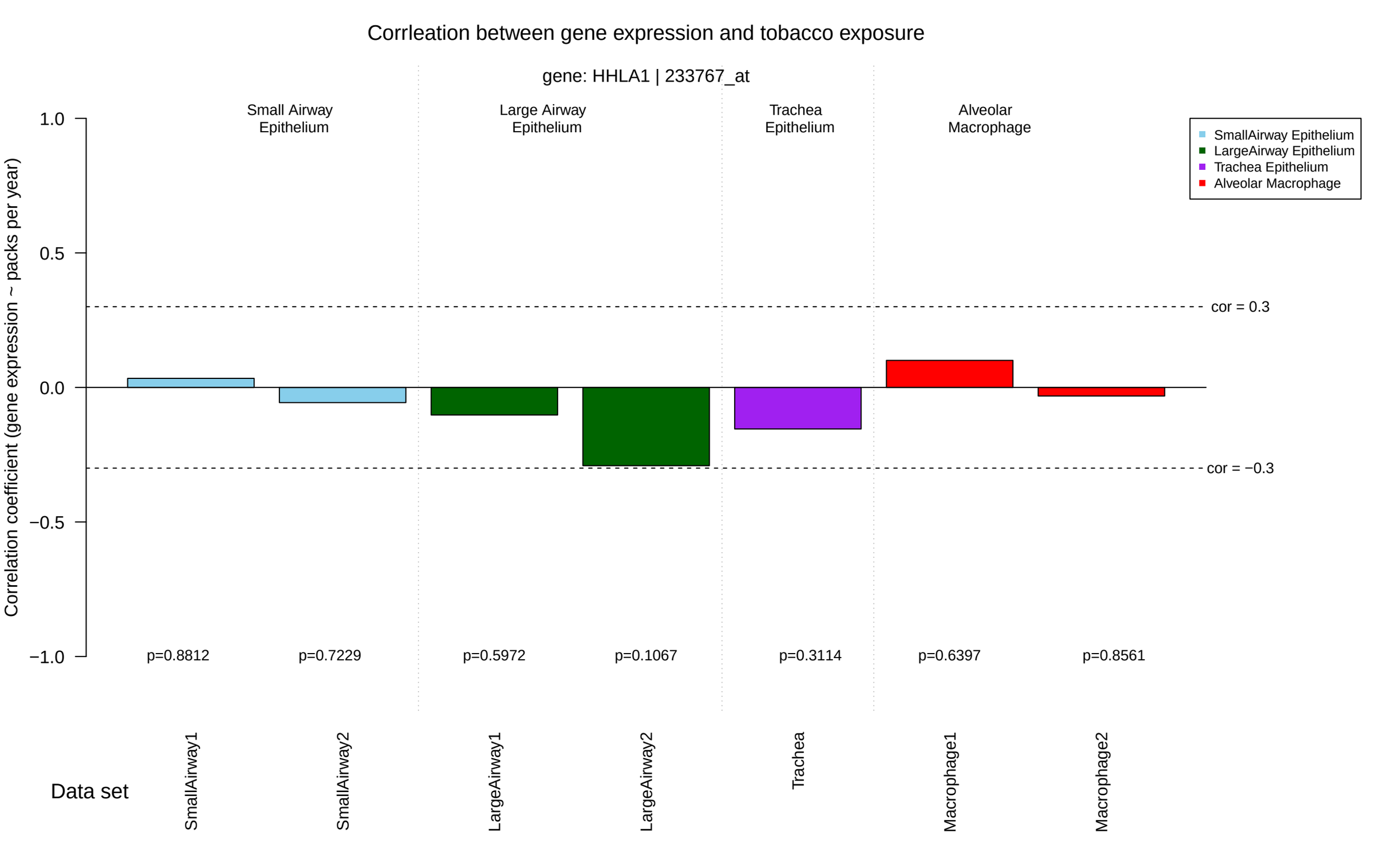

set: LargeAirway1 (p=0.477)
HHLA1 | 233767_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6329)
HHLA1 | 233767_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.3746)
HHLA1 | 233767_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0684)
HHLA1 | 233767_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.501)
HHLA1 | 233767_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HHLA1 | 233767_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8812
p=0.7229
p=0.5972
p=0.1067
p=0.3114
p=0.6397
p=0.8561
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2